Moles and Molar Quantities Unit: Sample Test

Name: _____ Pd: _____

Bring your own periodic table and calculator. Show your setup with units for every calculation. Part I is worth 1 point per question (20 points). Part II is on the following pages (46 points).

Part I: Multiple Choice

1. Which statement about one mole is correct?

A. One mole of any element has a mass of 1.00 g.
B. One mole is 6.022×10^{23} particles.
C. One mole of any substance occupies 22.4 L.
D. One mole of any compound contains 6.022×10^{23} atoms.

2. How many zinc atoms are in 0.250 mol of Zn?

A. 4.15×10^{-25} atoms
B. 1.51×10^{24} atoms
C. 6.022×10^{23} atoms
D. 1.51×10^{23} atoms

3. What is the molar mass of $Ca(NO_3)_2$?

A. 164.10 g/mol
B. 102.09 g/mol
C. 116.10 g/mol
D. 148.10 g/mol

4. How many moles are in 3.011×10^{23} molecules of CO_2?

A. 2.000 mol
B. 6.84×10^{21} mol
C. 0.5000 mol
D. 1.81×10^{47} mol

5. What is the mass of 2.50 mol of NaCl?

A. 0.0428 g
B. 146 g
C. 60.9 g
D. 1.51×10^{24} g

6. How many moles are in 88.0 g of CO_2?

A. 2.00 mol
B. 3870 mol
C. 0.500 mol
D. 3.93 mol

7. Which sample contains the greatest number of atoms?

A. 1.0 mol of He
B. 1.0 mol of H_2
C. 1.0 mol of N_2
D. 1.0 mol of CH_4

8. A student wants to convert a mass of a compound in grams into a number of molecules. What is the first step?

A. Multiply by 6.022×10^{23}.
B. Divide by the molar mass.
C. Divide by 22.4 L/mol.
D. Multiply by the molar mass.

9. What volume does 0.500 mol of N_2 gas occupy at STP?

A. 44.8 L
B. 314 L
C. 11.2 L
D. 0.0223 L

10. A gas sample occupies 33.6 L at STP. How many moles of gas is this?

A. 0.667 mol
B. 753 mol
C. 56.0 mol
D. 1.50 mol

11. Which 10.0 g gas sample has the greatest volume at STP?

A. H_2
B. He
C. N_2
D. CO_2

12. What is the percent by mass of oxygen in H_2O?

A. 11.21%
B. 50.00%
C. 88.79%
D. 94.06%

13. What is the percent by mass of nitrogen in NH_4NO_3?

A. 35.00%
B. 17.5%
C. 59.96%
D. 5.05%

14. What is the empirical formula of $C_6H_{12}O_6$?

A. $C_6H_{12}O_6$
B. CH_2O
C. $C_2H_4O_2$
D. CHO

15. **A compound is 75.0% C and 25.0% H by mass. What is its empirical formula?**
    A. CH_3
    B. C_3H_{12}
    C. CH_4
    D. C_2H_6

16. **A compound has the empirical formula CH_2O and a molar mass of 60.06 g/mol. What is its molecular formula?**
    A. CH_2O
    B. $C_3H_6O_3$
    C. $C_4H_8O_4$
    D. $C_2H_4O_2$

17. **What is the correct name for $CuSO_4 \cdot 5H_2O$?**
    A. copper(II) sulfate pentahydrate
    B. copper(I) sulfate pentahydrate
    C. copper(II) sulfate pentoxide
    D. copper(II) sulfate hydrate five

18. **A 5.00 g sample of a hydrate is heated until its mass is constant at 3.20 g. What is the percent water in the hydrate?**
    A. 64.0%
    B. 36.0%
    C. 56.3%
    D. 1.80%

19. **A student heats $CuSO_4 \cdot xH_2O$ and finds 0.0100 mol $CuSO_4$ and 0.0498 mol H_2O. What is x?**
    A. 4
    B. 6
    C. 5
    D. 0.0498

20. **1.00 mol of He and 1.00 mol of Ne contain the same number of atoms but have different masses. Why?**
    A. A mole of Ne contains more atoms than a mole of He.
    B. The mole is defined by mass, not by number.
    C. Ne atoms are always larger than He molecules.
    D. Each Ne atom has a greater mass than each He atom.

Part II: Free Response

1) Calculate each quantity. Show your setup with units and give the answer with the correct sig figs. (12 pts)

a. Find the molar mass of $(NH_4)_2CO_3$. (2 pts)

b. How many grams are in 0.750 mol of $Mg(OH)_2$? (2 pts)

c. How many molecules are in 25.0 g of glucose, $C_6H_{12}O_6$? (3 pts)

d. How many oxygen atoms are in 0.400 mol of $Al_2(SO_4)_3$? (3 pts)

e. What is the mass of 3.011×10^{23} molecules of NH_3? (2 pts)

2) Gases at STP. Use 22.4 L/mol where it applies. (10 pts)

a. What volume does 14.2 g of Cl_2 gas occupy at STP? (3 pts)

b. What is the mass of 5.60 L of NH_3 gas at STP? (3 pts)

c. How many molecules are in 11.2 L of N_2 gas at STP? (2 pts)

d. A student uses 22.4 L/mol for a gas at 25 °C and 1 atm. Explain why the answer will be wrong. (2 pts)

3) Percent composition and formulas. (12 pts)

a. Find the percent by mass of each element in $Al_2(SO_4)_3$. (4 pts)

b. A compound is 40.00% C, 6.71% H and 53.29% O by mass and has a molar mass of 180 g/mol. Find its empirical formula and its molecular formula. Show your work. (8 pts)

4) Hydrates. A student heats a sample of blue copper(II) sulfate hydrate, $CuSO_4 \cdot xH_2O$, in a crucible until the mass stops changing. (12 pts)

Data: mass of empty crucible = 18.42 g; crucible + hydrate (before heating) = 20.92 g; crucible + solid (after heating) = 20.02 g

a. Find the mass of the hydrate, the mass of water lost and the mass of anhydrous $CuSO_4$. (2 pts)

b. Calculate the moles of water lost and the moles of $CuSO_4$ left. (3 pts)

c. Find x. Then write the formula and the name of the hydrate. (3 pts)

d. Find the percent water by mass in this hydrate. (2 pts)

e. The student stopped heating early, so some water stayed in the crucible. Will the calculated x be too high or too low? Explain. (2 pts)